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Artificial assistant system notifications

Abstract

Artificial assistant system notification techniques are described that overcome the challenges of conventional search techniques. In one example, a user profile is generated to describe aspects of products or services learned through natural language conversations between a user and an artificial assistant system. These aspects may include price as well as non-price aspects such as color, texture, material, and so forth. To learn the aspects, the artificial assistant system may leverage spoken utterances and text initiated by the user as well as learn the aspects from digital images output as part of the conversation. Once generated, the user profile is then usable by the artificial assistant system to assist in subsequent searches.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. patent application Ser. No. 16/241,644, filed Jan. 7, 2019, entitled “Artificial Assistant System Notifications”, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application 62/614,899, filed Jan. 8, 2018, and entitled “Artificial Assistant System Notifications,” the entire disclosures of which are hereby incorporated by reference.

BACKGROUND

(1) Search is one of the primary techniques, via which, users of computing devices may locate information of interest. Users, for instance, may enter search queries to find digital content such as digital images, video, music, locate products or services, and so on. This is typically performed as a keyword search in which a user expresses a goal of the search to the best of their ability using text. This text is then matched to items of digital content that are tagged using similar keywords.

(2) Conventional search techniques, however, are typically repetitive and as a result may involve performance of repeated user interactions. A user, for instance, may initiate a search by a computing device for an item of interest. If that item is not found, the user is then forced to manually repeat the search. This may be due to unavailability of the item of interest or due to a

disconnect between how a goal of a search is expressed by a user using text and how the item of interest is tagged. Forced repetition as part of these conventional systems may thus hinder operation of the computing devices that implement the search functionality and result in user frustration.

SUMMARY

(3) Artificial assistant system notification techniques are described that overcome the challenges of conventional search techniques. In one example, a user profile is generated to describe aspects of products or services learned through natural language conversations between a user and an artificial assistant system. These aspects may include price as well as non-price aspects such as color, texture, material, and so forth. To learn the aspects, the artificial assistant system may leverage spoken utterances and text initiated by the user as well as learn the aspects from digital images output as part of the conversation, e.g., to learn colors, objects, and so on in digital images output by the artificial assistant system. The artificial assistant system may also learn values for these aspects over time that resulted in successful conversion of a product or service, e.g., to learn that a five percent discount is not interesting to a user but a ten percent discount did result in conversion.

(4) Once generated, the user profile is then usable by the artificial assistant system to assist in subsequent searches. The artificial assistant system, for instance, may detect user interest in a product or service as part of a natural language conversation. The artificial assistant system may then determine whether a triggering event has occurred with respect to that product or service (e.g., a product or service is available as having the aspect) based on the user profile, and if so, output a notification. In this way, the artificial assistant system may avoid the repeated manual searches of conventional techniques and also address limitations of an ability of a user to express a goal of a search through use of aspects learned from digital images.

(5) This Summary introduces a selection of concepts in a simplified form that are further described below in the Detailed Description. As such, this Summary is not intended to identify essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is described with reference to the accompanying figures. Entities represented in the figures may be indicative of one or more entities and thus reference may be made interchangeably to single or plural forms of the entities in the discussion.

(2) FIG. 1 is an illustration of an environment in an example implementation that is operable to employ artificial assistant system notification techniques described herein.

(3) FIGS. 2A-2C depicts an example of a natural-language conversation and notifications regarding a change in an aspect of a search.

(4) FIGS. 3-4 depict an example showing operation of an artificial assistant system of FIG. 1 in greater detail as supporting notification techniques described herein.

(5) FIG. 5 is a flow diagram depicting a procedure in an example implementation in which a user profile is generated and used as part of artificial assistant system notifications.

(6) FIG. 6 depicts an example of a real time camera platform usable to incorporate the artificial assistant system notification techniques described herein.

(7) FIG. 7 depicts a procedure in an example implementation in which a user profile is used to include additional aspects in a search to increase a likelihood of looking a product or service of interest.

(8) FIG. 8 illustrates an example system including various components of an example device that can be implemented as any type of computing device as described and/or utilize with reference to

FIGS. 1-7 to implement embodiments of the techniques described herein.

DETAILED DESCRIPTION

(9) Overview

(10) Conventional search techniques may involve repeated attempts to locate a desired product or service due to unavailability of the item of interest. This may also be due to a disconnect between how a goal of a search is expressed by a user using text and how the item of interest is tagged in order to locate the item. This results in computational inefficiency as well as user frustration.

(11) Accordingly, artificial assistant system notification techniques are described in which aspects of products or services learned through natural language conversations between a user and an artificial assistant system are used to generate a user profile. The user profile may then be leveraged in a variety of ways, such as to determine whether the aspect has been met by a product or service at a later point in time, automatically supplement subsequent searches, and so on. As a result, the artificial assistant system may support increased richness in searches of products and services with increased computational efficiency.

(12) In one example, the artificial assistant system detects user interest in a product or service. This may be detected in a variety of ways, such as whether a user “favorited” an item, as part of a natural language conversation, a search for the item, and so forth. A user, for instance, may engage in a natural language conversation with the artificial assistant system as part of an unsuccessful search to locate a product or service of interest. A user profile is generated by the artificial assistant system that identifies the user, the item, and an aspect of the product or service, which may include price and non-price aspects such as color, material, size, and so forth.

(13) The artificial assistant system then detects occurrence of a triggering event, which may be scheduled or based on the aspect itself such as a change in price a change availability of the non-price aspect, and so forth. In response, a search query is generated by the artificial assistant system automatically and without user intervention and a search result received in response to the search query. Digital images may also be leveraged, whether as part of the search query and/or to learn non-price aspects that are desired by the user, e.g., particular colors, textures, patterns, materials and so on. The artificial assistant system then determines whether the aspect is met, e.g., a price decrease by at least a threshold amount, availability of a particular color, size, and so forth. If so, the artificial assistant system outputs a notification indicating so to the user. In this way, the artificial assistant system may take proactive actions on the part of the user “in the background” to locate products or services of interest.

(14) The user profile may also be leveraged by the artificial assistant system to learn the aspects and values for the aspects over time. The artificial assistant system, for instance, may learn an amount of a discount that causes a user to purchase a product or service. This may also include non-price aspects, such as colors and so forth. These aspects may then be leveraged to supplement future searches. For example, the artificial assistant system may detect user interest in a product or service as before, e.g., as a result of an unsuccessful search. The artificial assistant system may then use an aspect learned in previous natural language conversations and maintained in a user profile to perform additional searches in the background to see if that aspect has been met and if so output a notification. A user, for instance, may search for running shoes but does not locate a running shoe of interest. The artificial assistant system may perform additional searches in the background, and as part of this also leverage the user profile to locate a particular shade of blue learned from previous successful conversions on the part of the user. As a result, a notification output by the artificial assistant system has an increased likelihood of being accurate and relevant to the user. Further discussion of these and other examples is included in the following sections and shown in corresponding figures.

(15) In the following discussion, an example environment is first described that may employ the techniques described herein. Example procedures are then described which may be performed in the example environment as well as other environments. Consequently, performance of the

example procedures is not limited to the example environment and the example environment is not limited to performance of the example procedures.

(16) Example Environment

(17) FIG. 1 is an illustration of a digital medium environment **100** in an example implementation that is operable to employ artificial assistant system notification techniques described herein. The illustrated environment **100** includes a service provider system **102** and a client device **104** of a user **106** that are communicatively coupled, one to another, via a network **108**. Computing devices that implement the service provider system **102** and the client device **104** may be configured in a variety of ways.

(18) A computing device, for instance, may be configured as a desktop computer, a laptop computer, a mobile device (e.g., assuming a handheld configuration such as a tablet or mobile phone as illustrated for the client device **104**), and so forth. Thus, a computing device may range from full resource devices with substantial memory and processor resources (e.g., personal computers, game consoles) to a low-resource device with limited memory and/or processing resources (e.g., mobile devices). Additionally, a computing device may be representative of a plurality of different devices, such as multiple servers utilized by a business to perform operations “over the cloud” as illustrated for the service provider system **102** and as described in relation to FIG. 8.

(19) The service provider system **102** in this example includes an artificial assistant system **110** that is configured to engage in a natural-language conversation **112** with a user **106** of the client device **104**. The client device **104**, for instance, may include an input device **114** that is usable to generate input data **116**, e.g., text via a keyboard, a spoken utterance that is converted to text, capture of a digital image, and so forth. The input data **116** is then communicated by a communication module **118** (e.g., browser, network-enabled application) via the network **108** to a conversation user interface **120** of the artificial assistant system **110**.

(20) The conversation user interface **120**, for instance, may be configured by an assistant interface module **122** and assistant manager module **124** to implement a chatbot, also known as an interactive agent or artificial conversational entity. The chatbot is configured to virtually simulate a conversational partner to a user **106** of the client device **104** as part of a dialog system to implement the natural-language conversation **112**. The assistant manager module **124**, for instance, may participate as part of the natural-language conversation **112** by generating communications as part of a “back-and-forth” with the user **106** of the client device **104** as shown in FIGS. 2A-2C. This may be employed in a variety of different scenarios, such as customer service, information acquisition, digital commerce, and so forth.

(21) To generate the communications, the assistant manager module **124** may employ a variety of functionality. In one example, analytics and rule-based logic **126** are employed, such as to parse the natural-language conversation **112** for keywords and to locate pre-configured replies based on the keywords. In another example, a machine-learning module **128** is employed that includes a natural-language processing module **130**. The machine-learning module **128**, for instance, may employ a model that is a computer representation that can be tuned (e.g., trained) based on inputs to approximate unknown functions.

(22) In particular, the model and machine-learning module **128** may employ computer functionality to learn from, and make predictions on, known data by analyzing the known data to learn to generate outputs that reflect patterns and attributes of the known data. Examples of machine learning include decision trees, support vector machines, linear regression, logistic regression, Bayesian networks, random forest learning, dimensionality reduction algorithms, boosting algorithms, artificial neural networks, deep learning, and so forth. Thus, machine learning and a machine learning model are used to make high-level abstractions in data by generating data-driven predictions or decisions from the known input data.

(23) Natural-language processing as implemented by the natural-language processing module **130**

is configured to incorporate speech recognition, natural-language understanding, and natural-language generation. This is used to understand communications received from the client device **104** as part of the natural-language conversation **112** as well as to generate communications to the client device **104**. The natural-language processing module **130**, for instance, may employ a neural network to generate a representation of a communication from the client device **104**, and from this, understand “what is being said” as well as to generate an appropriate response, if any. This may be used for a variety of purposes.

(24) In the illustrated example, the artificial assistant system **110** is configured to help control output of digital content **132** by the digital content control module **134** to the client device **104**, which is illustrated as stored in a storage device **136**. The digital content **132**, for instance, may be configured as recommendations, digital marketing content configured to cause conversion of a product or service, streaming digital media, digital music, search results, and so forth. Examples of this are utilized in search by a search module **138**, such as to locate a particular item of interest (e.g., a particular product or service), generate a search result, and so forth. A commerce module **140**, for instance, may be configured to receive bids for particular goods or services. Therefore, to locate an item of interest the search module **138** receives a search query from the artificial assistant system **110** and provides digital content **132** as a result of the search, e.g., identifying a particular item of interest.

(25) As previously described, conventional search techniques may involve significant amounts of repetition. Accordingly, techniques described herein are configured to reduce this repetition, and thus promote efficiency consumption of computational resources as part of performing a search. Examples **200**, **220**, **240** of this are illustrated in FIGS. 2A, 2B, and 2C.

(26) As depicting in the example **200** of FIG. 2A, a first natural language conversation **202** occurs between a user **106** of a client device **104** and the artificial assistant system **110** of the service provider system **102**, which may also be implemented locally by the client device **104**. In this conversation, the user **106** initiates a search query for a new game console and the artificial assistant system **110** provides a search result. A user **106** then provides an input indicating that an aspect of the search result is not satisfactory, e.g., price in this example. This may be understood by the artificial assistant system **110** through natural language processing and natural language understanding techniques, by noting that the product or service was not purchased, and so forth. Other non-price aspects are also contemplated, such as any characteristic of a product or service such as color, texture, material, size, pattern, style, and so forth.

(27) From this, the artificial assistant system **110** is configured to generate a user profile based on the user's expressed interest in a product or service that is subject to digital content retrieved as part of the search result as well as the identified aspect that is deemed unsatisfactory, e.g., price, size, color, texture, material, and so forth. The artificial assistant system **110** is then configured to repeat the search automatically and without user intervention based on a triggering event. A triggering event may be scheduled and/or based on a change to the aspect, e.g., price.

(28) As shown in the example **220** of FIG. 2B, for instance, a second natural language conversation **222** is initiated by the artificial assistant system **110** which include a notification of the change in the aspect, e.g., a drop-in price. In this instance, however, that change has not been deemed satisfactory by the user **106**. Accordingly, as shown in the example **240** of FIG. 2C, a third natural language conversation **242** is initiated by the artificial assistant system **110** indicative of a further change to the aspect. The user **106** in this instance deems this change agreeable, and in response provides an input to initiate purchase of the product or service. In this way, the artificial assistant system **110** may continue to operate “in the background” to initiate searches automatically and without user intervention.

(29) Further, the artificial assistant system **110** may be configured to learn preferences of the user **106** over time (e.g., as part of machine learning or rules and logic) with respect to these aspects (e.g., particular values for the aspects), a system to do which is illustrated with respect to FIGS. 3

and **4**. In the previous example, the artificial assistant system **110** may determine an amount of a change in the aspect that is deemed desirable by the user **106**, e.g., a dollar amount or percentage. This may then be employed by the artificial assistant system **110** in the future to control output of subsequent notifications regarding changes to these aspects.

(30) In general, functionality, features, and concepts described in relation to the examples above and below may be employed in the context of the example procedures described in this section. Further, functionality, features, and concepts described in relation to different figures and examples in this document may be interchanged among one another and are not limited to implementation in the context of a particular figure or procedure. Moreover, blocks associated with different representative procedures and corresponding figures herein may be applied together and/or combined in different ways. Thus, individual functionality, features, and concepts described in relation to different example environments, devices, components, figures, and procedures herein may be used in any suitable combinations and are not limited to the particular combinations represented by the enumerated examples in this description.

(31) FIG. 5 depicts a procedure **500** in an example implementation of artificial assistant system notifications. To begin in this example, user interest in a product or service (block **502**) is detected by a computing device. A user input, for instance, may be received via a user interface in which a user has selected an option to indicate a favorable impression of the product or service, e.g., has “favorited” the product or service. This may also be performed based on digital images communicated as part of the natural language conversation. For example, digital image processing techniques may be used to determine aspects of products or services that resulted in conversion and values of those aspects, e.g., colors, materials, patterns, objects, and so on. In another example, rules-based logic and/or machine learning is employed to monitor this user interaction, e.g., navigation to digital content (e.g., webpages) pertaining to particular products or services, repeated mention in a natural language conversation, purchase history, and/or other monitored user interaction with digital content.

(32) From this, a user profile is generated by the computing device that includes an identifier of the user (e.g., user ID), an identifier of the product or service (e.g., product ID), and an aspect of the product or service (block **504**), e.g., an aspect that was not “met” as previously described such as price, color, and so on. In one example, the aspect is stored automatically as a current price of the product or service as indicated by digital content received as part of a previous search result.

(33) The computing device then detects occurrence of a triggering event (block **506**). The triggering event, for instance, may be scheduled to occur at particular intervals of time, e.g., hourly, daily, weekly, monthly, quarterly, etc. In another example, the triggering event is defined based on a detected change to an aspect, e.g., “price” to find a better deal, availability of non-price aspects such as color, material, size, and so forth.

(34) In response to this detected occurrence, a search query is generated by the computing device (block **508**). Continuing with the previous example, a user may have “favorited” digital content that describes a product or service. A digital image included as part of this digital content is then used as part of the search query, which may also include characteristics of the product or service learned from the digital image, i.e., the aspects. Object recognition, for instance, may be performed by the computing device to determine what object is included in the digital image as well as aspects (e.g., characteristics) of the object, e.g., color, texture, size, and so forth. Thus, the digital image may provide additional information usable to increase accuracy of a search for aspects that may be difficult if not impossible for the user **106** to express using text, e.g., to describe a pattern.

(35) A search result is then received in response to a search performed using the search query (block **510**), e.g., using the digital image and accompanying data such as aspects learned from the image, text entered by the user, and so forth. This search may be performed remotely by a service provider system **102** and/or locally by the client device **104**. The search result includes digital content that results from the search, e.g., pictures, product or service listings, digital images, and so

forth.

(36) A determination is then made by the computing device as to whether the aspect is met by the search result (block **512**). For a price aspect, for instance, this may involve determining if the product or service is available for less. This may also be based on machine learning to determine a likelihood that this change is of a sufficient amount “to make a difference” to the user, e.g., by learning an amount of a price drop that will cause the user **106** to purchase a product or service. Based on this, a notification is output in response to the determination that the aspect is met (block **514**). The notification may be output in a variety of ways, such as by initiating another natural language conversation as shown in FIGS. 2A-2C, a message (e.g., instance message, text, email), served as digital marketing content to the user (e.g., a banner ad for output in a browser of mobile application), and/or as augmented or virtual reality content as further described in the following.

(37) FIG. 6 depicts an example system **600** a real time camera platform as incorporating notification techniques. To begin, a digital image **602** is obtained by digital camera **604** of a camera platform manager module **606**. The digital image **602**, for instance, may be captured using a digital camera, as a screenshot captured from a frame buffer of a computing device, and so forth. This may be performed, for instance, as part of a natural language conversation to supplement spoken utterances of the user **106**, as part of an initial search query, and so forth.

(38) The digital image **602** is then processed by an object recognition module **608** to recognize object aspects **614** exemplified by the digital image **602**. The object recognition module **608**, for instance, may employ a machine learning module **610** configured to employ models **612** usable to recognize the object and aspects of the object using machine learning, e.g., neural networks, convolutional neural networks, deep learning networks, structured vector machines, decision trees, and so forth.

(39) The models **612**, for instance, may be trained using training digital images that are tagged with corresponding identifications. In an implementation, these training digital images and tags are obtained from a commerce service provider system that are tagged by sellers using the system. As a result, a multitude of accurately tagged training digital images may be obtained with minimal computation and user cost as opposed to conventional manual tagging techniques. In this way, the models **612** may be configured to recognize aspects of objects that are usable to configure the user model to initiate search queries as described above. Although illustrated as implemented locally by the client device **104**, this functionality may also be implemented in whole or in part by a service provider system **102** via the network **108**.

(40) The object aspects **614** and digital image **602** are then communicated to the service provider system **102**. The service provider system **102** then uses the data as input data to perform a search, e.g., generate a search query and initiate a search. A result of the search performed based on the search query may then be output in real time in a user interface, e.g., as augmented or virtual reality content within a “live stream” of digital images captured in an environment of the client device **104**.

(41) The AR digital content, for instance, may include a name and price (e.g., average price, price for sale, price to buy, etc.) of object available via an electronic commerce system. The AR digital content, as rendered, is configured to remain “as is” relative to the view of the object, e.g., based on data received from orientation sensors such as accelerometers, inertial devices, gravimeters, image processing and feature recognition from the digital images, and so forth. The AR digital content, for instance, may indicate availability of a product or service that meets the aspect that was not previously met in an unsuccessful search.

(42) The AR digital content also includes information regarding the same or similar goods that are available for purchase from the service provider system **102**, e.g., as part of an online auction, for a set price, etc. In this way, the camera platform manager module **606** implements the camera platform as non-modal within a user interface such that a user remains within a context of a real time “live” feed of digital images and yet still is able to obtain metadata describing objects included

in those digital images. The user **106**, for instance, may “look around” at different objects within a living room and readily determine how to buy or sell these objects based on real time information obtained from the service provider system **102**. The user may also be made aware, via notifications output as AR digital content, of availability of products or services that meet aspects that were not previously met in previous searches. In this way, the artificial assistant system may expand functionality available to a user with increased computational efficiency over conventional techniques that require manual search repetition.

(43) FIG. 7 depicts a procedure **700** in an example implementation in which a user profile is used to include additional aspects in a search to increase a likelihood of looking a product or service of interest. A user profile is generated that describes products and services that are a subject of natural language conversations between an artificial assistant system and a user. The user profile identifies an aspect that resulted in successful conversion of a product or service as part of the natural language conversation (block **702**). A user as illustrated in FIGS. 2A-2C, for instance, may engage in repeated conversations with the artificial assistant system. These conversations may include spoken utterances, text inputs, and digital images as part of a back-and-forth conversation.

(44) A user, for instance, may engage in several natural language conversations. These conversations may involve different products and services and corresponding aspects, such as colors, discounts, materials, price, and so forth. Over time, the user profile is generated to “learn” which aspects results in successful conversion of a corresponding product or service, which may then be used to assist further searches. This may include analysis of text used as part of the natural language conversation (e.g., converted via spoken utterance) as part of natural language understanding, and may also leverage digital images used as part of the conversation. A user, for instance, may initiate a search for “blue shoes.” Digital image processing techniques may then be used to determine which shade of blue in a digital image resulted in successful conversion, e.g., purchase of the shoes. This aspect may then be added to the user profile to assist future searches.

(45) The artificial assistant system, for instance, may identify a search query involving at least one subsequent product or service is not successful (block **704**), e.g., to purchase gloves. A subsequent search query is then initiated by the artificial assistant system automatically and without user intervention for the subsequent product or service (block **706**), e.g., after a predefined amount of time. A determination is then made as to whether a triggering event defined with respect to the identified aspect of the user profile has occurred with respect to a search result resulting from the subsequent search query (block **708**). The artificial assistant system, for instance, may determine whether aspects specified by the user profile are met by the search result, e.g., a shade of blue that was not met in the unsuccessful search query. In this way, the artificial assistant system may locate products of services that may be of interest to a user from the unsuccessful searches, even if those aspects are not included in the original unsuccessful search. In an implementation, the subsequent search query includes this aspect from the user profile to increase a likelihood of inclusion of this aspect. A variety of other examples are also contemplated.

(46) Example System and Device

(47) FIG. 8 illustrates an example system generally at **800** that includes an example computing device **802** that is representative of one or more computing systems and/or devices that may implement the various techniques described herein. This is illustrated through inclusion of the artificial assistant system **110**. The computing device **802** may be, for example, a server of a service provider, a device associated with a client (e.g., a client device), an on-chip system, and/or any other suitable computing device or computing system.

(48) The example computing device **802** as illustrated includes a processing system **804**, one or more computer-readable media **806**, and one or more I/O interface **808** that are communicatively coupled, one to another. Although not shown, the computing device **802** may further include a system bus or other data and command transfer system that couples the various components, one to another. A system bus can include any one or combination of different bus structures, such as a

memory bus or memory controller, a peripheral bus, a universal serial bus, and/or a processor or local bus that utilizes any of a variety of bus architectures. A variety of other examples are also contemplated, such as control and data lines.

(49) The processing system **804** is representative of functionality to perform one or more operations using hardware. Accordingly, the processing system **804** is illustrated as including hardware element **810** that may be configured as processors, functional blocks, and so forth. This may include implementation in hardware as an application specific integrated circuit or other logic device formed using one or more semiconductors. The hardware elements **810** are not limited by the materials from which they are formed or the processing mechanisms employed therein. For example, processors may be comprised of semiconductor(s) and/or transistors (e.g., electronic integrated circuits (ICs)). In such a context, processor-executable instructions may be electronically-executable instructions.

(50) The computer-readable storage media **806** is illustrated as including memory/storage **812**. The memory/storage **812** represents memory/storage capacity associated with one or more computer-readable media. The memory/storage **812** may include volatile media (such as random access memory (RAM)) and/or nonvolatile media (such as read only memory (ROM), Flash memory, optical disks, magnetic disks, and so forth). The memory/storage **812** may include fixed media (e.g., RAM, ROM, a fixed hard drive, and so on) as well as removable media (e.g., Flash memory, a removable hard drive, an optical disc, and so forth). The computer-readable media **806** may be configured in a variety of other ways as further described below.

(51) Input/output interface(s) **808** are representative of functionality to allow a user to enter commands and information to computing device **802**, and also allow information to be presented to the user and/or other components or devices using various input/output devices. Examples of input devices include a keyboard, a cursor control device (e.g., a mouse), a microphone, a scanner, touch functionality (e.g., capacitive or other sensors that are configured to detect physical touch), a camera (e.g., which may employ visible or non-visible wavelengths such as infrared frequencies to recognize movement as gestures that do not involve touch), and so forth. Examples of output devices include a display device (e.g., a monitor or projector), speakers, a printer, a network card, tactile-response device, and so forth. Thus, the computing device **802** may be configured in a variety of ways as further described below to support user interaction.

(52) Various techniques may be described herein in the general context of software, hardware elements, or program modules. Generally, such modules include routines, programs, objects, elements, components, data structures, and so forth that perform particular tasks or implement particular abstract data types. The terms “module,” “functionality,” and “component” as used herein generally represent software, firmware, hardware, or a combination thereof. The features of the techniques described herein are platform-independent, meaning that the techniques may be implemented on a variety of commercial computing platforms having a variety of processors.

(53) An implementation of the described modules and techniques may be stored on or transmitted across some form of computer-readable media. The computer-readable media may include a variety of media that may be accessed by the computing device **802**. By way of example, and not limitation, computer-readable media may include “computer-readable storage media” and “computer-readable signal media.”

(54) “Computer-readable storage media” may refer to media and/or devices that enable persistent and/or non-transitory storage of information in contrast to mere signal transmission, carrier waves, or signals per se. Thus, computer-readable storage media refers to non-signal bearing media. The computer-readable storage media includes hardware such as volatile and non-volatile, removable and non-removable media and/or storage devices implemented in a method or technology suitable for storage of information such as computer readable instructions, data structures, program modules, logic elements/circuits, or other data. Examples of computer-readable storage media may include, but are not limited to, RAM, ROM, EEPROM, flash memory or other memory technology,

CD-ROM, digital versatile disks (DVD) or other optical storage, hard disks, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other storage device, tangible media, or article of manufacture suitable to store the desired information and which may be accessed by a computer.

(55) “Computer-readable signal media” may refer to a signal-bearing medium that is configured to transmit instructions to the hardware of the computing device **802**, such as via a network. Signal media typically may embody computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as carrier waves, data signals, or other transport mechanism. Signal media also include any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared, and other wireless media.

(56) As previously described, hardware elements **810** and computer-readable media **806** are representative of modules, programmable device logic and/or fixed device logic implemented in a hardware form that may be employed in some embodiments to implement at least some aspects of the techniques described herein, such as to perform one or more instructions. Hardware may include components of an integrated circuit or on-chip system, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), and other implementations in silicon or other hardware. In this context, hardware may operate as a processing device that performs program tasks defined by instructions and/or logic embodied by the hardware as well as a hardware utilized to store instructions for execution, e.g., the computer-readable storage media described previously.

(57) Combinations of the foregoing may also be employed to implement various techniques described herein. Accordingly, software, hardware, or executable modules may be implemented as one or more instructions and/or logic embodied on some form of computer-readable storage media and/or by one or more hardware elements **810**. The computing device **802** may be configured to implement particular instructions and/or functions corresponding to the software and/or hardware modules. Accordingly, implementation of a module that is executable by the computing device **802** as software may be achieved at least partially in hardware, e.g., through use of computer-readable storage media and/or hardware elements **810** of the processing system **804**. The instructions and/or functions may be executable/operable by one or more articles of manufacture (for example, one or more computing devices **802** and/or processing systems **804**) to implement techniques, modules, and examples described herein.

(58) The techniques described herein may be supported by various configurations of the computing device **802** and are not limited to the specific examples of the techniques described herein. This functionality may also be implemented all or in part through use of a distributed system, such as over a “cloud” **814** via a platform **816** as described below.

(59) The cloud **814** includes and/or is representative of a platform **816** for resources **818**. The platform **816** abstracts underlying functionality of hardware (e.g., servers) and software resources of the cloud **814**. The resources **818** may include applications and/or data that can be utilized while computer processing is executed on servers that are remote from the computing device **802**. Resources **818** can also include services provided over the Internet and/or through a subscriber network, such as a cellular or Wi-Fi network.

(60) The platform **816** may abstract resources and functions to connect the computing device **802** with other computing devices. The platform **816** may also serve to abstract scaling of resources to provide a corresponding level of scale to encountered demand for the resources **818** that are implemented via the platform **816**. Accordingly, in an interconnected device embodiment, implementation of functionality described herein may be distributed throughout the system **800**. For example, the functionality may be implemented in part on the computing device **802** as well as

via the platform **816** that abstracts the functionality of the cloud **814**.

CONCLUSION

(61) Although the invention has been described in language specific to structural features and/or methodological acts, it is to be understood that the invention defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed invention.

Claims

1. A method for outputting a notification by an artificial assistant system of a computing device, the method comprising: generating, by the computing device, a user profile including an aspect model trained using machine learning based on initial natural language conversations between the artificial assistant system and a user, the user profile being stored in memory and identifying a first visual aspect that resulted in successful conversion of a first product or service as part of the initial natural language conversations; identifying, by the computing device, a second visual aspect that resulted in unsuccessful conversion of a second product or service as part of a subsequent natural language conversation, the user profile being updated with the second visual aspect; learning, automatically and as a result of the unsuccessful conversion by the computing device and using the aspect model trained using machine learning, the second visual aspect by analyzing previous natural language conversations without requiring explicit user input about visual preferences; initiating, by the computing device and using the user profile, a search query based on the first visual aspect and the second visual aspect; repeating, automatically and without user intervention by the computing device in a background of the computing device, one or more searches using the search query until a determination that a triggering event has occurred; and outputting, by the artificial assistant system, a notification responsive to the determination that the triggering event has occurred.

2. The method as described in claim 1, wherein the search query is for the second product or service, and the triggering event corresponds to the second product or service having at least one of the first visual aspect or the learned second visual aspect being available via an electronic commerce system.

3. The method as described in claim 1, wherein the search query is for a different product or service, and the triggering event corresponds to the different product or service having at least one of the first visual aspect or the learned second visual aspect being available via an electronic commerce system.

4. The method as described in claim 1, wherein the triggering event is scheduled to occur at particular intervals of time.

5. The method as described in claim 1, wherein the first visual aspect is at least one of texture, material, and pattern, and the learned second visual aspect is color, the first visual aspect or the learned second visual aspect being learned as an analysis of text and digital images used in the previous natural language conversations.

6. The method as described in claim 1, wherein the second visual aspect includes a particular shade of color that is learned based on one or more of the initial language conversations that resulted in successful conversion of products or services without requiring explicit user input.

7. The method as described in claim 1, wherein the first visual aspect and the second visual aspect are identified from digital images output by the artificial assistant system as part of the initial and the subsequent natural language conversations, respectively.

8. The method as described in claim 1, wherein the first visual aspect and the second visual aspect are identified from digital images captured by a camera of a client device of the user and provided as input to the artificial assistant system as part of the initial and the subsequent natural language conversations, respectively.

9. The method as described in claim 8, wherein the first visual aspect and the second visual aspect are identified using an object recognition module trained to identify object aspects in the digital images using machine learning based on tagged training images obtained from an electronic commerce system.

10. The method as described in claim 8, wherein the digital images are captured by the client device as part of a live stream of an environment of the client device, and the outputting the notification includes outputting the notification as augmented reality digital content within the live stream.

11. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform operations comprising: generating a user profile including an aspect model trained using machine learning based on natural language conversations between an artificial assistant system of a computing device and a user, the user profile being stored in memory and identifying a first visual aspect that resulted in successful conversion of a first product or service and a second visual aspect that resulted in unsuccessful conversion of a second product or service; learning, automatically and as a result of the unsuccessful conversion and using the aspect model trained using machine learning, the second visual aspect by analyzing previous natural language conversations without requiring explicit user input about visual preferences, the user profile being updated with the second visual aspect; repeating, automatically and without user intervention and in a background of the computing device, one or more searches based on the first visual aspect and the second visual aspect from the user profile until a determination that a triggering event has occurred; and outputting, by the artificial assistant system, a notification responsive to determining that a triggering event has occurred.

12. The non-transitory computer-readable medium of claim 11, wherein the triggering event is based on availability of the second product or service having at least one of the first visual aspect or the learned second visual aspect via an electronic commerce system.

13. The non-transitory computer-readable medium of claim 11, wherein the triggering event is scheduled to occur at particular intervals of time.

14. The non-transitory computer-readable medium of claim 11, wherein the second visual aspect includes a particular shade of color that is learned based on the previous natural language conversations that resulted in successful conversion of products or services without requiring explicit user input.

15. The non-transitory computer-readable medium of claim 11, wherein the first visual aspect and the second visual aspect are identified based on an analysis of text and digital images output by the artificial assistant system as part of the natural language conversations.

16. The non-transitory computer-readable medium of claim 11, wherein the first visual aspect and the second visual aspect are identified from digital images captured by a camera of a client device of the user and provided as input to the artificial assistant system as part of the natural language conversations.

17. A computing device, comprising: a processing system; and a memory to store instructions which, responsive to execution by the processing system, cause the processing system to perform operations including: detecting user interest in a product or service and a visual aspect that resulted in unsuccessful conversion of the product or service as part of a natural language conversation between an artificial assistant system of the computing device and a user; learning, automatically and as a result of the unsuccessful conversion and using an aspect model trained using machine learning, the visual aspect based on an analysis of previous natural language conversations that resulted in successful conversion of different products or services without requiring explicit user input about visual preferences; generating a user profile that includes the learned visual aspect and at least one additional visual aspect that resulted in successful conversion of an additional product or service as part of the previous natural language conversations, the user profile being stored in the memory; repeating, automatically and without user intervention and in a background of the

computing device, one or more searches based on the product or service, the user profile, the learned visual aspect, and the at least one additional visual aspect until a triggering event occurs; and outputting, by the artificial assistant system, a notification responsive to a determination that the triggering event has occurred.

18. The computing device of claim 17, wherein the triggering event corresponds to the product or service having at least one of the learned visual aspect or the at least one additional visual aspect being available via an electronic commerce system.

19. The computing device of claim 17, wherein the visual aspect is color, and the learned visual aspect is a particular shade of color.

20. The computing device of claim 17, wherein the visual aspect is identified from an analysis of text and digital images included as part of the previous natural language conversation.
